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ME1100 Thermodynamics

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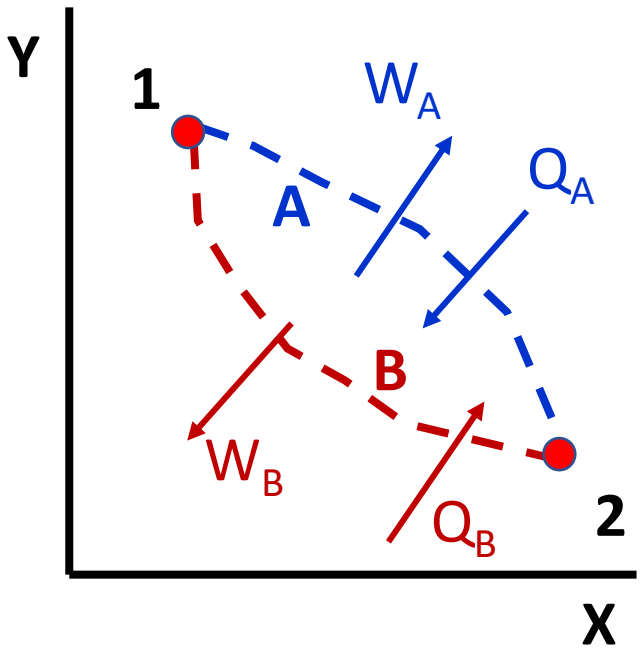
Work & Heat: Recapitulation

Sign convention

Work: +ve when done BY the system, and,

Heat flow: +ve when added TO the system

Work and heat flow are path functions



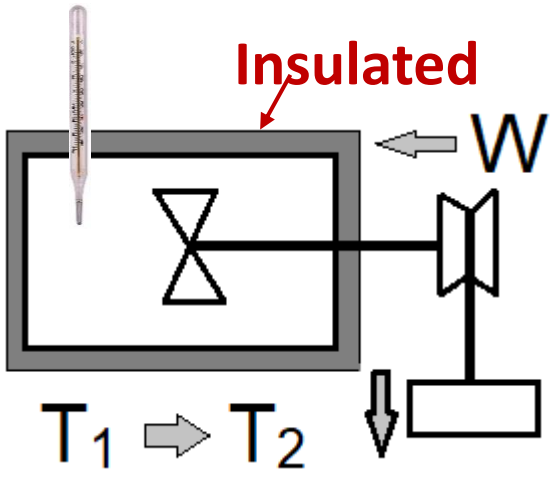
Change in property X (or, Y) = X_2-X_1 (or, Y_2-Y_1)

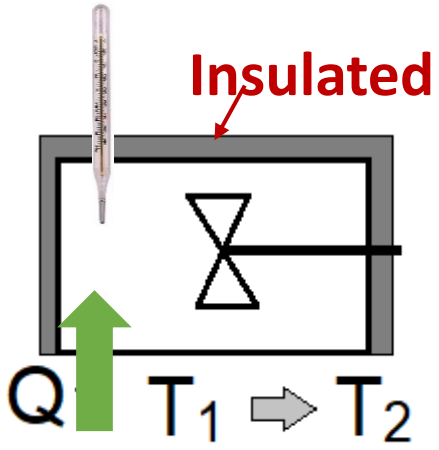
$$_1W_2 \neq W_2 - W_1 \qquad _1Q_2 \neq Q_2 - Q_1$$

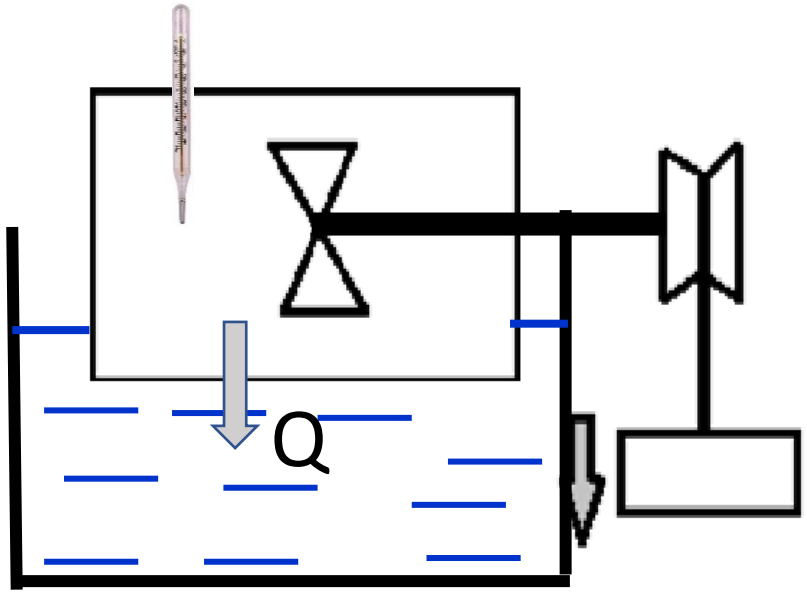
First law of thermodynamics

- Conservation of energy
- Conversion of heat to work / work to heat

Joule's Experiment



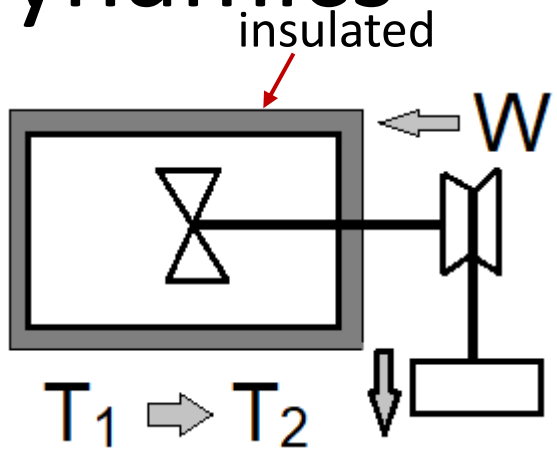




Same temperature rise in both cases

First law of thermodynamics

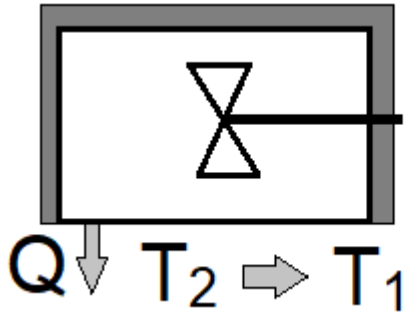
Joule's Experiment



Consider a system undergoing a cyclic process (at least two different processes being present, the system returns to its same initial state).

In process 1 – 2, paddle work was done on an insulated system by lowering an object of mass m through a distance, Temperature increased from T_1 to T_2.

In process 2 – 1 heat was transferred out of the system to reinstate the temperature from T_2 to T_1. No paddle work was done here.



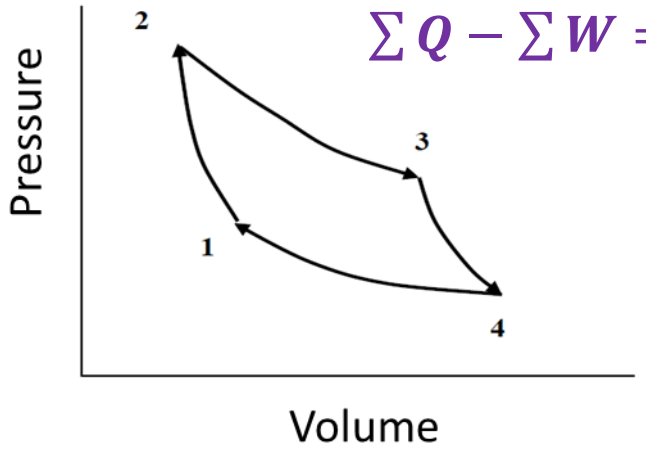
He repeated the experiments with several working fluids, in different containers, at different times.

$$_1W_2 = {}_2Q_1$$

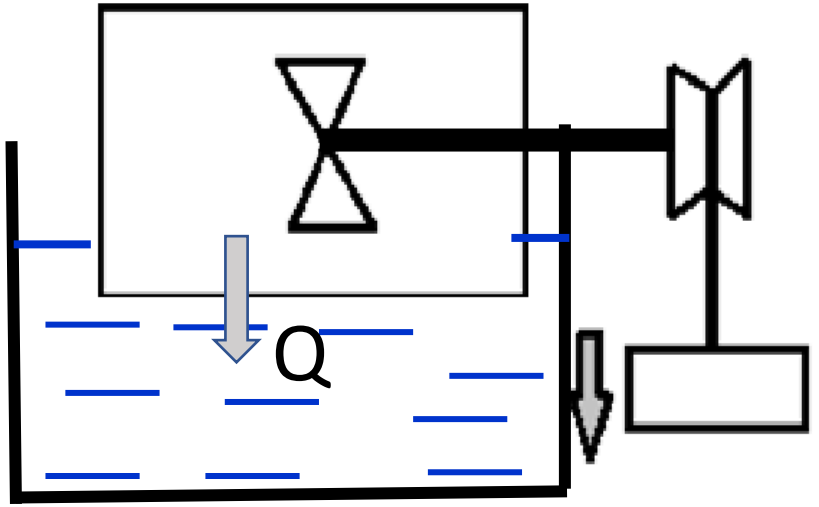
It was found that the value of Q in Joule was equal to that of W in Joule.

First law of Thermodynamics

Thus for a general cyclic process,

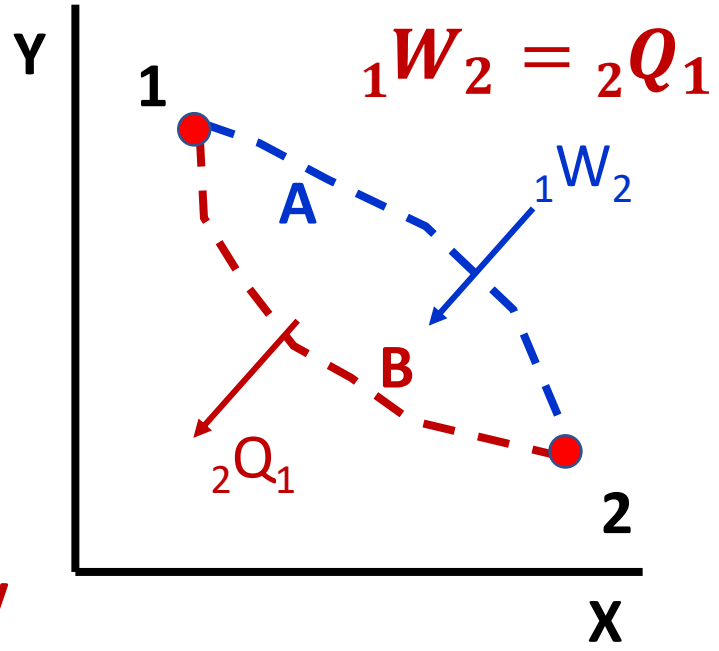


$$\sum Q - \sum W = 0 \text{ or } \oint (\delta Q - \delta W) = 0$$



Algebraic sum of all work and heat interactions between a system and its surroundings in a cyclic process is zero.

$$_1W_2 + {}_2W_3 + {}_3W_4 + {}_4W_1 = {}_1Q_2 + {}_2Q_3 + {}_3Q_4 + {}_4Q_1 \Rightarrow \oint (\delta Q - \delta W) = 0$$



The cyclic integral of a point function is zero and property is a point function

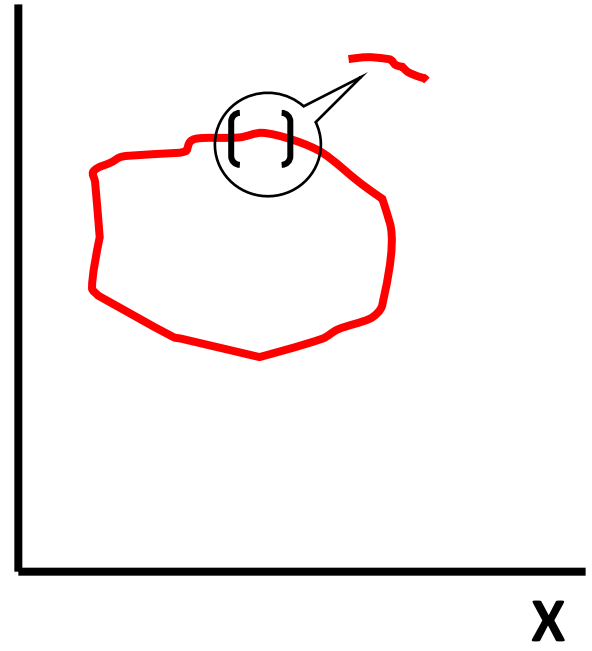
$$\oint d\chi = 0$$

χ is any property

First law of Thermodynamics

$\delta Q - \delta W = dE$

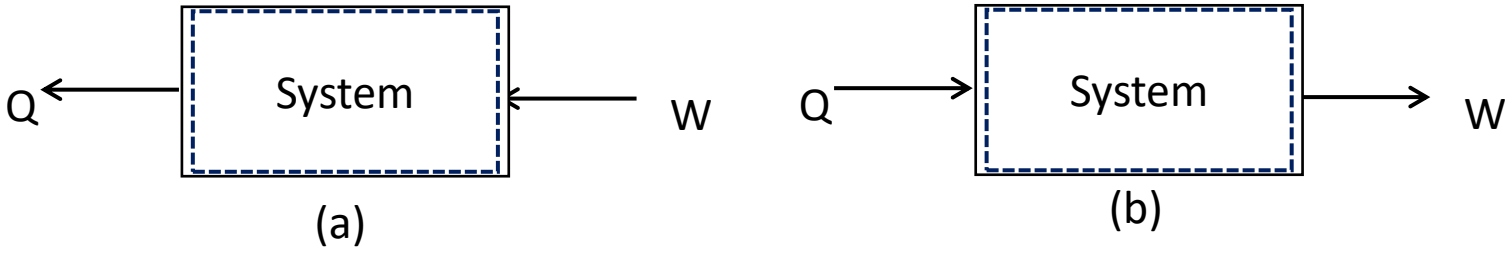
Difference between heat and work interaction for an infinitesimally small process



For a finite process (non-cyclic)

$Q - W = \Delta E$

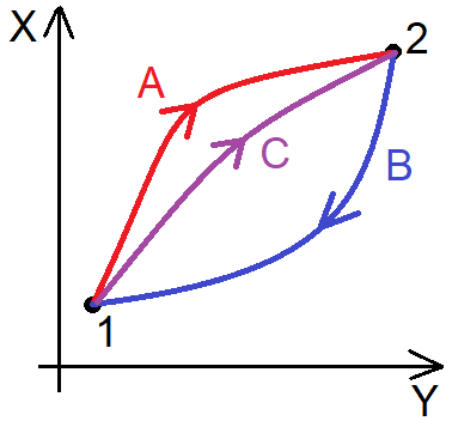
- **Heat is path function, Work is path function**
- **But their difference is a state/point function**



(a) W=-10kJ, Q=-6kJ **ΔE=4kJ**

(b) W=6kJ, Q=10kJ **ΔE=4kJ**

In cases where input energy interaction is less than output, results in energy release from the system



Consider two independent processes
1-A-2-B-1 and 1-C-2-B-1

First law for process 1-A-2-B-1 :

$$Q_{1-A-2} + Q_{2-B-1} = W_{1-A-2} + W_{2-B-1} \quad (1)$$

First law for process 1-C-2-B-1 :

$$Q_{1-C-2} + Q_{2-B-1} = W_{1-C-2} + W_{2-B-1} \quad (2)$$

- Subtraction of (2) from (1) and a slight rearrangement gives:

$$Q_{1-A-2} - W_{1-A-2} = Q_{1-C-2} - W_{1-C-2} \quad (3)$$

- Notice that we did not make any assumptions regarding the two processes 1-A-2 and 1-C-2. They are arbitrary processes.

- Since equation (3) is valid for both the paths, it is evident that the difference of Q and W is independent of path, and function of end states only

- Difference is **stored energy (E)**

Stored energy E

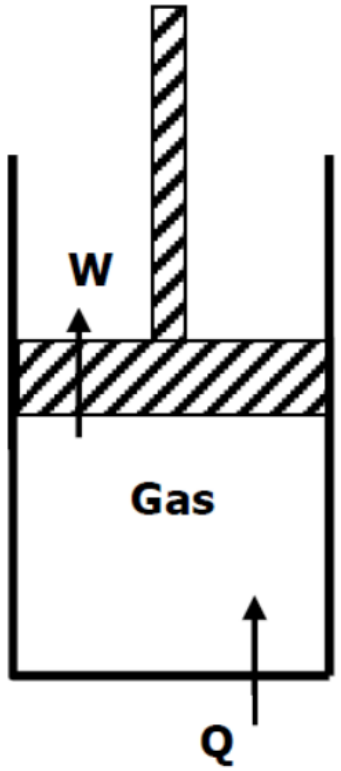
- Stored energy (E) is sum of different forms of energy
  - Kinetic energy, Potential energy, electrical energy, magnetic energy, internal energy
- $\Delta E = \Delta KE + \Delta PE + \Delta EE + \Delta ME + \Delta U$
- The properties associated with the individual energy forms (at fixed mass) are:
  - KE (kinetic energy) - velocity;
  - PE (potential energy)- elevation;
  - EE- charge; ME- magnetization;
  - Internal energy (U) – Temperature

Internal energy (denoted U) is the energy possessed by the molecules that comprise the system in the form of translational and rotational kinetic energies.

First law of Thermodynamics for system with ONLY displacement work

- For a system which does not have energy storage in the form of kinetic energy, potential energy, electrical energy, etc. the first law can be simplified to the form

$$\boldsymbol{Q - W = \Delta U}$$



- For an infinitesimal change, the above equation can be rewritten as

$$\boldsymbol{\delta Q - \delta W = \mathrm{d}U}$$

- Substituting for the expansion work ($\boldsymbol{\delta W}$) of the gas as **pdV**

$$\boldsymbol{\delta Q = \mathrm{d}U + p\,dV}$$

- We write the infinitesimal heat added as δq since **heat is a path function** (not a property, inexact differential) and write internal energy change as dU **since internal energy is a property** (exact differential)

  - Note that U is total internal energy (Joules).

  - We now define a **new property called enthalpy H** as follows

$$H = U + PV$$

  - **The first law for a non-cyclic process can now be written as:**

$$\delta Q = dH - V\,dP$$

Specific heat

- **Specific heat at constant pressure** is defined as:

$$c_p = \left(\frac{\partial h}{\partial T}\right)_p$$

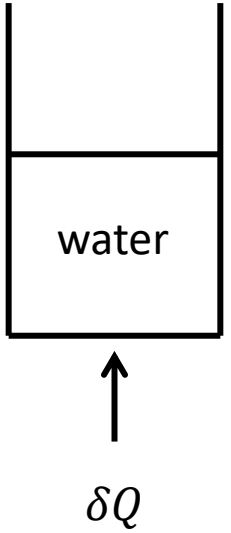
- Similarly, **specific heat at constant volume** is defined as:

$$c_v = \left(\frac{\partial u}{\partial T}\right)_v$$

- **In high school, you have learnt c_p in terms of heat added as the amount of heat required to raise the temperature of a fluid of mass m by 1 K.**

$$\boldsymbol{c_p = \frac{\delta Q}{m\, dT}}$$

Specific heat



Since δQ = dH – 0 (pressure is constant)

$$c_p = \frac{dH}{m\,\mathrm{d}T} = \frac{dh}{dT} \quad \textit{at constant pressure}$$

$$c_p = \left(\frac{\partial h}{\partial T}\right)_P$$

Thank You